# Many-body expansion of the CT commutators

generated by `expand_commutators.py`

All commutator blocks of the `commutator` stage expanded with the generalized Wick theorem (`generalized_wick_theorem/generalized_wick_commutator.py`). The truncation is by perturbation order (Møller–Plesset counting $\hat{F} \sim O(1)$, $\hat{V} \sim O(\lambda)$, $\hat{T} \sim O(\lambda)$), correct through $O(\lambda^2)$: the first-order commutators $[\hat{F}, \hat{A}]$, $[\hat{V}, \hat{A}]$ and the Fock half of the second order, $\frac{1}{2}[[\hat{F}, \hat{A}], \hat{A}]$. The valence double commutator is $O(\lambda^3)$ and is not expanded. Repeated indices are summed over their space. The index spaces and the delta rules are those of `readme.md`:

$$p, q, r, s: \text{ whole space}, \qquad i, j, k, l: \text{ occupied}, \qquad a, b, c, d: \text{ virtual}, \tag{1}$$

$$\delta(\text{whole}, \text{vir}) \to \text{virtual}, \qquad \delta(\text{whole}, \text{occ}) \to \text{occupied}, \qquad \delta(\text{occ}, \text{vir}) = 0. \tag{2}$$

The operators are

$$\hat{F} = f^q_p \, \hat{E}^p_q, \qquad \hat{V} = \tfrac{1}{2} g^{pq}_{rs} \, \hat{E}^{rs}_{pq}, \qquad \hat{T}_1 = t^a_i \, \hat{E}^a_i, \qquad \hat{T}_2 = \tfrac{1}{2} t^{ab}_{ij} \, \hat{E}^{ab}_{ij}, \tag{3}$$

with $\hat{F}$, $\hat{V}$ Hermitian; the second excitation of the nested commutators carries the dummy indices $c, d$ and $k, l$. The coefficients are real and follow the spin-integrated convention: no index antisymmetry is used anywhere, $g^{pq}_{rs}$ and $t^{ab}_{ij}$ carry only the simultaneous pair swap $g^{pq}_{rs} = g^{qp}_{sr}$, $t^{ab}_{ij} = t^{ba}_{ji}$, and $f^q_p = f^p_q$, $g^{pq}_{rs} = g^{rs}_{pq}$. Conjugate blocks are expanded by conjugating the first-order result term by term ($\hat{E}^{u\cdots}_{l\cdots} \to \hat{E}^{l\cdots}_{u\cdots}$, $f^q_p \to f^p_q$, $g^{pq}_{rs} \to g^{rs}_{pq}$; $t$ is real and unchanged). Blocks containing operators of rank higher than two are marked; their cumulant expansion (and the $[\ldots]_{1,2}$ truncation) is done separately.

Equal terms are merged (`readme.md`, rule e) over every identity that is exact in this convention: renaming of any summed index within its space — the free indices of the operator are the summation labels of the operator basis sum, so they can be renamed like tensor dummies — the pair symmetries of $g$ and $t$ above, and the exact fermionic row antisymmetry of $\hat{E}$.

## 1 First-order commutators

$$\left[\hat{F}, \hat{T}_1\right] = -f^i_p t^a_i \hat{E}^a_p + f^a_p t^a_i \hat{E}^p_i \tag{4}$$

$$\left[\hat{F}, \hat{T}_2\right] = -f^j_p t^{ab}_{ij} \hat{E}^{ab}_{ip} + f^b_p t^{ab}_{ij} \hat{E}^{ap}_{ij} \tag{5}$$

$$\left[\hat{V}, \hat{T}_1\right] = -g^{ip}_{qr} t^a_i \hat{E}^{ap}_{qr} + g^{ar}_{pq} t^a_i \hat{E}^{pq}_{ir} \tag{6}$$

$$\left[\hat{V}, \hat{T}_2\right] = -\tfrac{1}{2} g^{ij}_{pq} t^{ab}_{ij} \hat{E}^{ab}_{pq} + \tfrac{1}{2} g^{ab}_{pq} t^{ab}_{ij} \hat{E}^{pq}_{ij} \tag{7}$$

$$- g^{jp}_{qr} t^{ab}_{ij} \hat{E}^{abp}_{iqr} + g^{br}_{pq} t^{ab}_{ij} \hat{E}^{apq}_{ijr} \quad \text{(three-body terms)} \tag{8}$$

## 2 Second order: Fock blocks

$$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_1\right] = -2\,f^b_j t^b_i t^a_j \hat{E}^a_i \tag{9}$$

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right] = f^i_j t^b_j t^a_i \hat{E}^a_b - f^b_p t^b_i t^a_i \hat{E}^a_p \tag{10}$$

$$+\, f^a_b t^a_i t^b_j \hat{E}^i_j - f^j_p t^a_j t^a_i \hat{E}^p_i \tag{11}$$

$$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_2\right] = f^c_k t^a_k t^{bc}_{ij} \hat{E}^{ab}_{ij} - f^c_k t^c_j t^{ab}_{ik} \hat{E}^{ab}_{ij} \tag{12}$$

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right] = -f^j_k t^c_k t^{ab}_{ij} \hat{E}^{ab}_{ci} \tag{13}$$

$$- f^c_p t^c_j t^{ab}_{ij} \hat{E}^{ab}_{ip} + f^b_c t^c_i t^{ab}_{jk} \hat{E}^{ai}_{jk} - f^k_p t^b_k t^{ab}_{ij} \hat{E}^{ap}_{ij} \tag{14}$$

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_1\right] = -f^c_k t^{ab}_{ik} t^c_j \hat{E}^{ab}_{ij} + f^c_k t^{bc}_{ij} t^a_k \hat{E}^{ab}_{ij} \tag{15}$$

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right] = -f^j_k t^{bc}_{ik} t^a_j \hat{E}^{ai}_{bc} \tag{16}$$

$$+ f^c_p t^{bc}_{ij} t^a_j \hat{E}^{ai}_{bp} - f^c_p t^{bc}_{ji} t^a_j \hat{E}^{ai}_{bp} - f^i_p t^{bc}_{ij} t^a_j \hat{E}^{ap}_{bc} \tag{17}$$

$$+ f^b_c t^{ab}_{ij} t^c_k \hat{E}^{ij}_{ak} - f^a_p t^{ab}_{ij} t^b_k \hat{E}^{ij}_{kp} - f^k_p t^{ab}_{ik} t^b_j \hat{E}^{ip}_{aj} \tag{18}$$

$$+ f^k_p t^{ab}_{ki} t^b_j \hat{E}^{ip}_{aj} \tag{19}$$

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right] = 2\,f^d_l t^{cd}_{jk} t^{ab}_{il} \hat{E}^{abc}_{ijk} \quad \text{(three-body terms)} \tag{20}$$

$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right] = f^i_j t^{cd}_{jk} t^{ab}_{ik} \hat{E}^{ab}_{cd} \tag{21}$$

$$- f^d_p t^{cd}_{ij} t^{ab}_{ij} \hat{E}^{ab}_{cp} + f^a_b t^{ac}_{ij} t^{bc}_{kl} \hat{E}^{ij}_{kl} - f^l_p t^{ab}_{il} t^{ab}_{jk} \hat{E}^{ip}_{jk} \tag{22}$$

$$- f^k_l t^{cd}_{il} t^{ab}_{jk} \hat{E}^{abi}_{cdj} - f^d_p t^{cd}_{ik} t^{ab}_{jk} \hat{E}^{abi}_{cjp} \tag{23}$$

$$+ f^d_p t^{cd}_{ki} t^{ab}_{jk} \hat{E}^{abi}_{cjp} - f^j_p t^{cd}_{jk} t^{ab}_{ik} \hat{E}^{abp}_{cdi} \tag{24}$$

$$- f^c_d t^{bd}_{ij} t^{ac}_{kl} \hat{E}^{aij}_{bkl} - f^b_p t^{bc}_{ij} t^{ac}_{kl} \hat{E}^{aij}_{klp} \tag{25}$$

$$+ f^l_p t^{bc}_{il} t^{ac}_{jk} \hat{E}^{aip}_{bjk} - f^l_p t^{bc}_{li} t^{ac}_{jk} \hat{E}^{aip}_{bjk} \quad \text{(three-body terms)} \tag{26}$$

## 3 Assembly

With the blocks above, the transcorrelated Hamiltonian through $O(\lambda^2)$ is

$$\left[\hat{F},\,\hat{A}\right] = \left[\hat{F},\,\hat{T}_1\right] + \left[\hat{F},\,\hat{T}_2\right] + \text{h.c.}, \tag{27}$$

$$\left[\hat{V},\,\hat{A}\right] = \left[\hat{V},\,\hat{T}_1\right] + \left[\hat{V},\,\hat{T}_2\right] + \text{h.c.}, \tag{28}$$

$$\tfrac{1}{2}\left[\left[\hat{F},\,\hat{A}\right],\,\hat{A}\right] = \tfrac{1}{2}\sum_{r,s=1}^{2}\left(\left[\left[\hat{F},\,\hat{T}_r\right],\,\hat{T}_s\right] + \left[\left[\hat{F},\,\hat{T}_r\right]^{\dagger},\,\hat{T}_s\right]\right) + \text{h.c.}, \tag{29}$$

the valence double commutator $\frac{1}{2}[[\hat{V},\hat{A}],\hat{A}] \sim O(\lambda^3)$ being dropped by the perturbation-order truncation.